

**EBN111 SEMESTER TOETS/TEST 1 (13 MAR 2014)****ANTWOORDBLAD / ANSWER SHEET**

|                         |  |                                    |           |  |         |  |  |  |
|-------------------------|--|------------------------------------|-----------|--|---------|--|--|--|
| Van / Surname:          |  | Studentenommer:<br>Student number: |           |  |         |  |  |  |
| Voorletters / Initials: |  | Taal / Language                    | Afrikaans |  | English |  |  |  |

|    |  |    |  |    |  |             |            |
|----|--|----|--|----|--|-------------|------------|
| Q1 |  | Q2 |  | Q3 |  | <b>Tot:</b> | <b>/45</b> |
|----|--|----|--|----|--|-------------|------------|

**VRAAG 1 – QUESTION 1 (7)**

|     |                           |                        |                        |              |
|-----|---------------------------|------------------------|------------------------|--------------|
| 1.1 | $a) p(1s) = -2 \text{ W}$ | $b) W = 432 \text{ J}$ | $c) R = 1.25$          | $d) Q = 5mC$ |
| 1.2 | $P_1 = 30.72 \text{ W}$   | $P_2 = 20 \text{ W}$   | $P_3 = -396 \text{ W}$ |              |

**VRAAG 2 – QUESTION 2 (16)**

|     |                                                      |                                |                               |
|-----|------------------------------------------------------|--------------------------------|-------------------------------|
| 2.1 | Onafhanklike lusse: 3<br><i>Independent loops: 3</i> | Takke: 9<br><i>Branches: 9</i> | Nodusse: 7<br><i>Nodes: 7</i> |
| 2.2 | $I = -5 \text{ A}$                                   | $v_o = -5 \text{ V}$           | $R = 1.875 \Omega$            |
| 2.3 | $v_y = 6 \text{ mV}$                                 |                                |                               |
| 2.4 | $I_1 = 2 \text{ mA}$                                 | $G_1 = 2.5 \mu S$              |                               |
| 2.5 | $R_{eq} = 10 \Omega$                                 |                                |                               |
| 2.6 | $a) R_{cb} = 42.55 \Omega$                           | $b) R_{ab} = 72 \Omega$        |                               |

**VRAAG 3 – QUESTION 3 (22)**

|     |                                                                                                                                       |          |                           |         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------|---------|
| 3.1 | $V_o = 12 \text{ V}$                                                                                                                  |          |                           |         |
| 3.2 | $i_X = 0.3 \text{ A}$                                                                                                                 |          |                           |         |
| 3.3 | $a = 3$                                                                                                                               | $b = -1$ | $c = -1$                  | $d = 2$ |
| 3.4 | $e = 40$                                                                                                                              | $f = 50$ | $g = 1$                   | $h = 0$ |
| 3.5 | $\begin{bmatrix} 20 & -12 \\ -12 & 40 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 7 \end{bmatrix}$ |          |                           |         |
| 3.6 | $a) V_{CE} = 3.2 \text{ V}$                                                                                                           |          | $b) V_S = 2.6 \text{ V}$  |         |
| 3.7 | $a) k = -6$                                                                                                                           |          | $b) V_2 = -0.5 \text{ V}$ |         |